

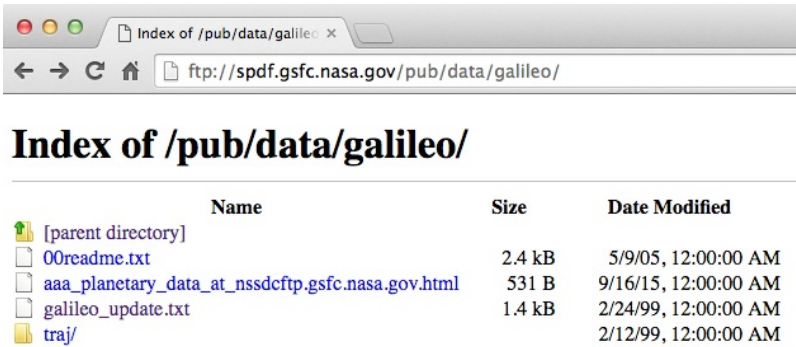
1.1 Web history

Internet and web

Computers of the 1940s and 1950s were mostly solitary machines, but in the 1960s computer networks evolved. A computer network allows computers to communicate information with each other. The **internet** began as four networked computers in 1969. Today, the internet is the largest and most popular computer network, spanning the entire globe.

Early versions of the internet looked and acted similar to today, but in a much simpler way. Documents were plain text: no images, no formatting, and no interaction. **FTP (File Transfer Protocol)** was an early way for transferring files over the internet. Users used FTP programs to connect to servers, look at listings of available documents, and download documents of interest.

Figure 1.1.1: NASA public FTP site.



In the early 1990s, Tim Berners-Lee was working at a Swiss research institute named CERN and developed a more convenient way for computers to communicate files over the internet. Berners-Lee named his creation the **World Wide Web**, or simply "the **web**". The web involved three things:

- 1. Text files, known as HTML files, containing *links* to other text files.
- 2. A program, known as a browser, for viewing HTML files.
- 3. A set of rules, known as the HTTP protocol, for transferring HTML files among computers.

PARTICIPATION
ACTIVITY

1.1.1: The web uses HTML files, web browsers, and HTTP to provide a more convenient way for computers to communicate on the web.

Animation content:

An FTP computer terminal is displayed on the left with the following commands: login, get file 1, get file 2.

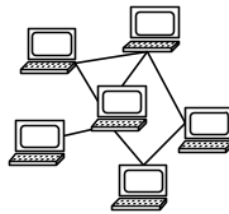
On the right, a travel blog is displayed with examples of links to files and how clicking on a link will open another file.

Animation captions:

- 1. FTP is used to transfer files by logging into a computer with the files and typing commands to get the desired files.
- 2. HTTP is a protocol for transferring HTML files among computers. Files can contain links to other files.
- 3. Clicking a link in a file opens the file being linked to. Files can contain multiple links.

The web was originally called the World Wide Web, or WWW, because Berners-Lee envisioned a large collection of globally distributed webpages linking to each other. A **webpage** or **web page** is a document that is viewed in a web browser. A collection of related webpages are organized into a **website**. A **web server** is a program that serves webpages to web browsers.

Figure 1.1.2: The web's name comes from the interconnections of computers being like the interconnections of a spider's web.



Source: Spider web ([Steve Gibson](#) / Public Domain), computer network (zyBooks)

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Introduction of HTML

HyperText markup language (HTML) is the standard markup language for web documents. **Hypertext** is text that has links to other text (and today to images, videos, and more). Document **markup** is special markings in the document that provide additional information about links, formatting, and images. HTML also permits adding metadata like search engine keywords, author information, and language.

Figure 1.1.3: One of the first webpages.

World Wide Web

The WorldWideWeb (W3) is a wide-area [hypermedia](#) information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an [executive summary](#) of the project, [Mailing lists](#), [Policy](#), November's [W3 news](#), [Frequently Asked Questions](#).

[What's out there?](#)

Pointers to the world's online information, [subjects](#), [W3 servers](#), etc.

[Help](#)

on the browser you are using

[Software Products](#)

A list of W3 project components and their current state. (e.g. [Line Mode](#), [X11 Viola](#), [NeXTStep](#), [Servers](#), [Tools](#), [Mail robot](#), [Library](#))

[Technical](#)

Details of protocols, formats, program internals etc

[Bibliography](#)

Paper documentation on W3 and references.

[People](#)

A list of some people involved in the project.

[History](#)

A summary of the history of the project.

[How can I help?](#)

If you would like to support the web..

[Getting code](#)

Getting the code by [anonymous FTP](#), etc.

Hypertext history

Tim Berners-Lee's ideas of creating linking webpages was founded on previous work on hypertext:

- 1945: Engineer Vannevar Bush writes the essay "As We May Think" that describes Memex, a theoretical machine for building and following links between documents.
- 1965: Ted Nelson coins the term HyperText in a paper on how to deal with information that was complex, changing, and uncertain.
- 1968: Doug Engelbart demonstrates an implementation of hyperlinks with a mouse in the oN-Line System (NLS).
- 1987: Apple releases HyperCard, software for the Macintosh that enables programming hypertext applications.

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PARTICIPATION
ACTIVITY

1.1.2: The web.



If unable to drag and drop, refresh the page.

browser

HTTP

Early 1990s

webpage

website

web server

HTML

	The period when the web was first developed
	The language used for a text file that describes a webpage.
	Web document that is viewed in a web browser.
	Collection of related webpages.
	A program that serves webpages to browsers.
	The protocol for transferring files among computers via the web.
	A program for viewing webpages.

Reset

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Web vs. internet

Today, much of the internet traffic is web traffic. People thus have a hard time distinguishing the internet (the interconnection of computers communicating using a set of rules), and the web (which is just one particular use of the internet). Besides transferring webpages from one computer to another, the internet also transmits email, music, video, and other types of data.

Table 1.1.1: Number of websites and internet users per year.

The number of websites has grown at a dramatically fast rate.

Year	Websites	Internet users	Major websites launched
2018	1,630,322,579	3,896,000,000	
2017	1,766,926,408	3,650,000,000	
2016	1,045,534,808	3,417,000,000	
2015	863,105,652	3,185,996,155	
2014	968,882,453	2,925,249,355	
2013	672,985,183	2,756,198,420	
2012	697,089,489	2,518,453,530	
2011	346,004,403	2,282,955,130	Twitch
2010	206,956,723	2,045,865,660	Pinterest, Instagram
2009	238,027,855	1,766,206,240	
2008	172,338,726	1,571,601,630	Dropbox
2007	121,892,559	1,373,327,790	Tumblr
2006	85,507,314	1,160,335,280	Twitter
2005	64,780,617	1,027,580,990	YouTube, Reddit
2004	51,611,646	910,060,180	Thefacebook, Flickr
2003	40,912,332	778,555,680	WordPress, LinkedIn
2002	38,760,373	662,663,600	
2001	29,254,370	500,609,240	Wikipedia

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2000	17,087,182	413,425,190	
1999	3,177,453	280,866,670	PayPal
1998	2,410,067	188,023,930	Google
1997	1,117,255	120,758,310	eBay, Netflix
1996	257,601	77,433,860	
1995	23,500	44,838,900	Altavista, Amazon
1994	2,738	25,454,590	Yahoo!
1993	130	14,161,570	
1992	10		
8/1/91	1		

Source: [InternetLiveStats, 2018](#), and [Statista](#)

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PARTICIPATION ACTIVITY

1.1.3: Websites per year.

Refer to the above table.

1) When was the first website published?

- ☐ 1991
☐ 2001

2) When was Yahoo created?

- ☐ 1994
☐ 2001

3) How many websites existed in 2001?

- ☐ Thousands
☐ 30 million

4) How many websites existed in 2011?

- ☐ 350 million
☐ 350 trillion

5) In what year did the number of internet users surpass 3 billion?

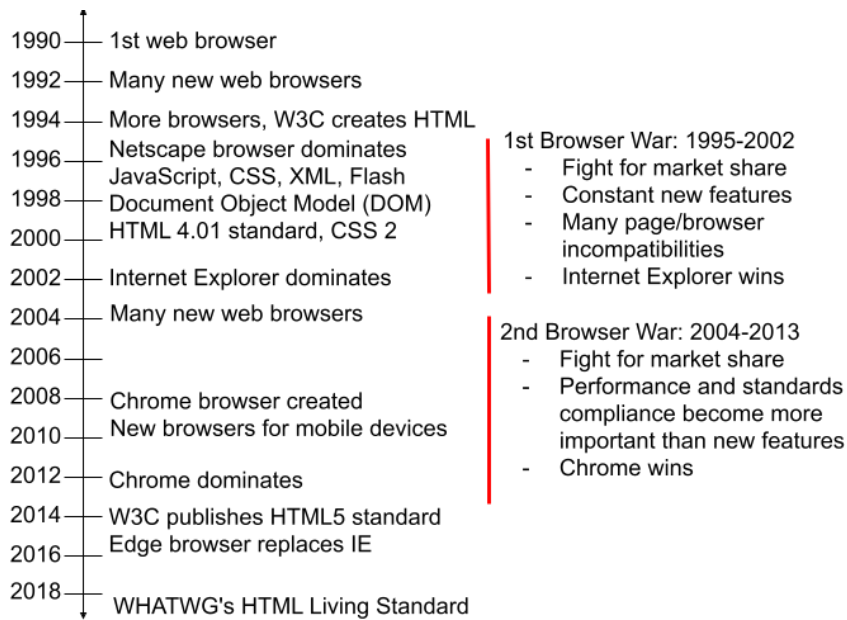
- ☐ 2005
☐ 2015

Browser wars and HTML standardization

A **web browser** is a program that downloads an HTML document from a web server, displays the document to the user with the appropriate formatting, and allows the user to interact with the document, such as clicking hyperlinks to access other documents. A web browser uses HTML to understand the structure and semantics, or meaning, of the document.

Early in browser history, browser developers competed for users by trying to provide the best web browsing experience. Ex: Browser developers added enhancements allowing greater interactivity in web documents. These enhancements only worked within specific browsers, so many documents could not be viewed properly on all browsers.

Figure 1.1.4: Web browser timeline.



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The frequent webpage and browser incompatibility headaches pushed the industry to value standardization. The **World Wide Web Consortium (W3C)** is the international standards organization that traditionally has controlled a number of web standards, including HTML. HTML5 was the latest HTML standard released by the W3C in 2014.

In 2019, the W3C relinquished HTML standards publishing to the **Web Hypertext Application Technology Working Group (WHATWG)**, an organization that develops a variety of web standards and whose members include the major browser vendors. WHATWG produces the **HTML Living Standard**, a continually evolving standard without version numbers that replaces HTML5.

A webpage that conforms to the HTML Living Standard will look and act the same way in most modern web browsers. With standardization, browser developers now compete on browser speed, standards compliance, and browser features rather than on the basis of proprietary extensions.

W3C vs. WHATWG

The W3C and WHATWG organizations have had a somewhat stormy relationship over the years. For some time, W3C and WHATWG even published competing HTML standards. However, the two organizations are now agreed that WHATWG will be setting the HTML standard. The term "HTML5" is likely to be synonymous with "HTML Living Standard" for some time.

PARTICIPATION ACTIVITY

1.1.4: Significant points in web history.

If unable to drag and drop, refresh the page.

1995

1965

2019

1990

2014

1945

1994

The idea for generating links from within one document to other documents.

The term HyperText invented.

First web browser is created.

The World Wide Web Consortium (W3C) created.

HTML 2.0 standard published.

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HTML5 standard published.

WHATWG controls the HTML standard.

Reset

Separation of duties

A significant change that occurred over time was a move to separate document structure, document presentation (how the document is displayed in a browser), and webpage interaction with the user. Document markup was initially used to control both document structure and appearance. Some markup, such as the tag `<b>`, was originally used just to control appearance. Interlacing document structure with presentation and interaction complicates having pages work well across the range of technologies from large screens, to small phones, to printing devices.

A modern webpage is composed of HTML, CSS, and JavaScript.

- HTML defines the structure and content of a webpage.
- CSS specifies the layout and visible appearance.
- JavaScript describes the dynamic behaviors and actions of a webpage.

PARTICIPATION ACTIVITY

1.1.5: Separation of structure, presentation, and interaction.

If unable to drag and drop, refresh the page.

CSS

JavaScript

HTML

Specifies document structure

Manages document presentation

Enables document interaction

Reset

Great example

[CSS Zen Garden](#) is a website where each page consists of exactly the same HTML and text, but different CSS is used to make the pages look totally different.

Exploring further:

- [Internet history timeline: ARPANET to the World Wide Web](#) (Live Science)
- [W3C.org](#)
- [The history of the Web](#) (W3C)
- [HTML Living Standard](#) (WHATWG)

1.2 IP addresses, domain names, and URLs

IP addresses

A computer communicates with another computer on the internet by sending packets back and forth. An internet **packet** contains To and From IP addresses, the information to communicate, and other configuration information.

An **IP address** (short for **Internet Protocol** address) is a computer's unique address on the internet (like a house's unique address in the world), usually represented numerically like 198.51.100.7. A typical IP address is 32 bits, divided into four 8-bit groups, each group often written as a decimal number.

PARTICIPATION ACTIVITY

1.2.1: IP addresses.

Animation content:

The following IP address is displayed: 198.51.100.7. For each section of the IP address, the corresponding 8 bits value is displayed. The 198 is represented with the 8 bits 11000110, 51 is 00110011, 100 is 01100100, and 7 is 00000111.

Animation captions:

1. An IP address is usually written as four numbers separated by dots.
2. Each number represents 8 bits of the IP address, for a total of 32 bits.
3. The smallest possible number in an IP address is 0, and the largest is 255.

The original Internet Protocol, known as **IPv4**, has 32-bit addresses. 32 bits can represent 2^{32} or about 4 billion unique addresses, originally believed to be more than enough, but 4 billion is no longer enough. A new version of the Internet Protocol, **IPv6**, uses 128-bit addresses, capable of representing 2^{128} addresses. The number 2^{128} is 3.4×10^{38} or 340,000,000,000,000,000,000,000,000,000,000,000,000,000, which is hopefully enough for quite a while. IPv4 and IPv6 currently co-exist and likely will for a long time.

PARTICIPATION ACTIVITY

1.2.2: IP addresses.

- 1) What is the size of an IP address in IPv4?
 - ☐ 32 bits
 - ☐ 128 bits
- 2) IPv6 supports about how many IP addresses?
 - ☐ 100 billion
 - ☐ 3.4×10^{38}
- 3) With the introduction of IPv6, devices on the internet no longer use IPv4.
 - ☐ True
 - ☐ False

Domain names and DNS

Some websites can be directly reached using the computer system's IP address. Ex: Google could say "Go to 216.58.193.206 to search the internet." But those numbers are hard to remember, and IP addresses can change, so domain names are commonly used. A **domain name** is a name for an IP address, such as the name wikipedia.org for the IP address 198.35.26.96; the name is easier to remember and type. Capitalization doesn't matter: Wikipedia.org, wikipedia.org, and WIKIPEDIA.ORG are treated the same.

When a computer sends a packet using a domain name over the internet, the first step is to contact a **DNS server** to convert the domain name to an IP address. **DNS** is short for Domain Name System.

Thirteen main DNS servers (called **root servers**) exist in the world, and a computer's operating system or an ISP keeps a reference to the root servers' IP addresses. Ex: 198.41.0.4 (run by Verisign), 192.228.79.201 (run by USC), 199.7.91.13 (run by Univ. of Maryland), and 192.203.230.10 (run by NASA). The first step of sending an internet packet to a domain name is thus to lookup the IP address via a DNS server.

PARTICIPATION ACTIVITY

1.2.3: A DNS server first looks up a domain name's IP address.

Animation content:

A computer is displayed with the following IP address: 198.51.100.0. A packet is displayed right next

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to the 198.51.100.0 computer with the information: To: cnn.com, From: 198.51.100.00, Type: Web Access, Data:. Then another packet is sent to a DNS server to get the IP address for cnn.com. This packet retrieves 157.166.226.25 under Data: and sends it to the original packet. Now the original packet has the IP address for cnn.com, which is 157.166.226.25.

Animation captions:

1. A computer wants to send a packet to cnn.com.
2. First, a packet is sent to a DNS server to lookup the IP address for cnn.com.
3. The packet contains the destination address. The computer can now communicate with CNN.

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Registering a domain name

Anyone may register an unused domain name with a **domain name registrar**. Most registrars charge a yearly fee for keeping the domain registered. Registration information is made publicly available from ICANN's [Whois](#) service.

When a website is hosted with a web hosting company, the company will update the DNS servers so the website's domain name is associated with the IP address of the hosted website. The web hosting company may also manage the yearly registration of the domain name.

PARTICIPATION ACTIVITY

1.2.4: IP addresses and domain names.

1) Websites have *either* an IP address or a domain name, but not both.

- ☐ True
☐ False

2) cnn.com is a(n) ____.

- ☐ IP address
☐ domain name
☐ DNS

3) A DNS server's primary role is to convert ____.

- ☐ an IP address into 32 bits
☐ an IP address to a domain name
☐ the domain name to an IP address

Try 1.2.1: IP addresses and domain names.

Several sites will lookup the IP address of a domain, such as the following.

- [Click to open www.site24x7.com](#). Type "Wikipedia.com" or your favorite website name.

Given an IP address, some sites will indicate the geographic location of that IP address' computer. Try entering 171.67.215.200, which is Stanford.com's IP address, whose computers are in Palo Alto, California.

- [Click to open www.iplocation.net](#).

Typing "What is my IP?" in a search engine can be used to determine one's IP address.

- [Click to open Google.com](#).
- [Click to open Bing.com](#).

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Domain name levels

Domain names are hierarchical. A domain name belongs to one of numerous **top-level domains (TLD)**, such as .com, .net, .org, .edu, and .gov. Also, each country is assigned a unique two-letter **country code top-level domain (ccTLD)** like .uk (United Kingdom), .ru (Russia), and .de (Germany). **ICANN**, the organization that manages TLDs, now allows companies and organizations to create customized TLDs, like .church, .pizza, and .music.

PARTICIPATION ACTIVITY

1.2.5: Common top-level domains.

Animation content:

A branch diagram is displayed with each branch explaining the domain name of websites. The Commercial websites are represented with a .com domain name, Network websites are represented with a .net domain name, Organization websites are represented with a .org domain name, Education websites are represented with a .edu domain name, and Government sites are represented with a .gov domain name.

Animation captions:

1. Internet domain names are divided by top-level domains.
2. The most popular is .com. Most companies have a .com name.
3. Next most popular is .net. Typically for networking/internet-focused companies.
4. Next is .org, traditionally for non-profit organizations.
5. Other well-known top-level domains are .edu for colleges, and .gov for U.S. governments. Hundreds more exist.

Immediately after a top-level domain comes a **second-level domain**, such as wikipedia in wikipedia.org. A second-level domain is commonly an organization's name as in Stanford.edu, or indicates the purpose of a website as in DoPython.org. Third-level and further level domains refer to sub-computer systems local to an organization, as in cs.stanford.edu where the cs is for Stanford's Computer Science department. A common third-or-deeper-level domain is www, short for World Wide Web, usually referring to an organization's web server. Many organizations use www optionally, so stanford.edu by default goes to www.stanford.edu.

PARTICIPATION ACTIVITY

1.2.6: Domain hierarchy.

Given a website such as www.whitehouse.gov, match the domain to the corresponding domain-level hierarchy.

If unable to drag and drop, refresh the page.

www

.gov

whitehouse

Top-level domain.

Third-level domain

Second-level domain.

Reset

Note

The top-down structure of a domain name is, somewhat counterintuitively, from right-to-left rather than left-to-right. Thus, cs.stanford.edu has a top-level domain of .edu, second-level stanford, and third-level cs. Common scams use this counterintuitive right-to-left feature to trick people into thinking a site belongs to a particular company. Ex: bankofamerica.xyz.com might trick a person into thinking the domain name is for Bank of America.

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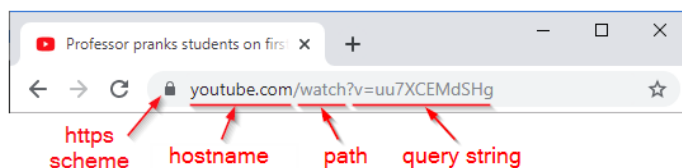
URLs

Domain names are most commonly seen in URLs. A **URL (Uniform Resource Locator)** is the location of a web resource on the web, such as <http://www.cdc.gov/alcohol/faqs.htm>. A **web resource** is any retrievable item, like an HTML file, image, video, CSS stylesheet, etc.

A URL is composed of several parts:

- **Scheme** - Characters at the beginning of a URL followed by a colon ":" or a colon and double slashes "://". Common URL schemes include **http**, **https**, **mailto**, and **file**. Ex: In <https://www.cdc.gov/alcohol>, the scheme is "https".
- **Hostname** - The complete domain name following the scheme in a URL. Ex: In <https://www.cdc.gov/alcohol>, the hostname is "www.cdc.gov".
- **Path** - The characters to the right of the hostname in a URL. Ex: In <https://www.cdc.gov/alcohol>, the path is "/alcohol".
- **Query string** - Optional characters to the right of the question mark (?) in a URL that provide data for the web server. In <https://www.youtube.com/watch?v=uu7XCEMdSHg>, the characters after the ? tells YouTube's server to play a video having code uu7XCEMdSHg.
- **Fragment** - Optional characters at the end of a URL that start with a hash character (#) and refer to a certain location within a webpage. In <https://en.wikipedia.org/wiki/URL#History>, the fragment "#History" refers to the webpage's History section.

Figure 1.2.1: Chrome browser hides the scheme and "www" in hostname.



PARTICIPATION ACTIVITY

1.2.7: A URL can specify the location of subfolders and files.

Animation content:

A computer sends a packet with the information To: <http://www.cdc.gov/alcohol/faqs.htm>, Type: Web access, Data: "FAQ" to a web server at www.cdc.gov. This packet retrieves and sends back information to the computer from a subfolder named alcohol and then another subfolder named faqs.htm.

Animation captions:

1. A computer's browser sends a packet requesting information from a web server at IP address 23.39.150.112, with hostname www.cdc.gov.
2. The information is in a subfolder named alcohol, then in a file in that folder named faqs.html.
3. The faqs.html file is returned and displayed in the computer's browser.

PARTICIPATION ACTIVITY

1.2.8: URLs.

1) Is the following a valid URL?

<https://www.whitehouse.gov/>

- ☐ Yes
☐ No

2) Is the following a valid URL?

https://www.nasa.gov/cubequest/details/#.VOaO_VPF9U9

- ☐ Yes
☐ No

3) What is the hostname of the following URL?

<http://www.weather.com/summer/temperatures.html>

- ☐ www.weather.com

☐ weather.com

4) Which file does the following URL access? □

http://www.weather.com/summer/temperatures.html

☐ summer

☐ temperatures.html

5) Which scheme should a URL use to provide a secured HTTP connection to a web server? □

☐ http

☐ https

6) Is www.nasa.gov a valid URL? □

☐ Yes

☐ No

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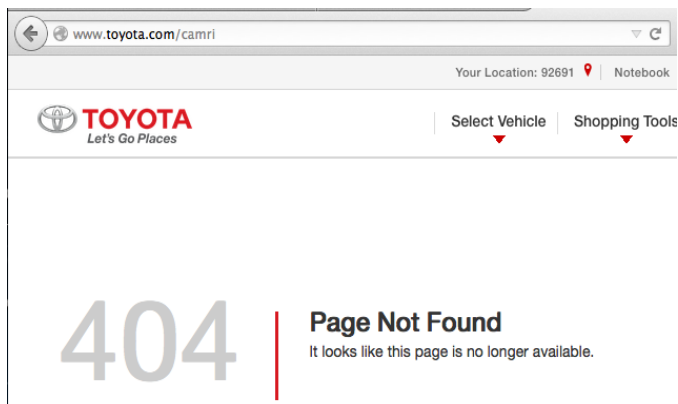
Webpage errors

If a domain name is not found by a DNS server (usually because the domain name is not registered), a page is displayed indicating such, as in "Sorry, the website www.xyz.blahblahblah cannot be found."

A domain name may be found (so a valid IP address exists for that name) but the web server may not respond, resulting in a message like "The website is not responding" or "Could not reach the website". Such non-response could be due to the web server being turned off or undergoing maintenance, or due to an essential router malfunctioning, for example.

If a web server is reached but the specific requested page isn't found, the server returns a **404** status code, which is a code number for page not found. Various other status codes exist. Many web servers return a page that includes the number "404".

Figure 1.2.2: An example webpage that returns a message with the code 404 shown.



Linkrot

The web is always changing with content being continually added and removed. When content is removed from the web, URLs that used to point to the content return a 404 status code. **Linkrot** is the general name for a once valid link that now returns a 404 status code.

The [Internet Archive](#) is one of many organizations around the world that fights linkrot by archiving the web for posterity. The Internet Archive's [Wayback Machine](#) provides access to billions of archived webpages and can often show users what a particular URL looked like at different times in history.

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1.2.9: Webpage errors.

- 1) A 404 message indicates that a domain name is not recognized.
- ☐ True
- ☐ False
- 2) If a web server is down (such as powered off), the server may return a page saying to please try again later.
- ☐ True
- ☐ False

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Exploring further:

- [URL Standard](#) (WHATWG)
- [List of Top-Level Domains](#) (ICANN)

CHALLENGE
ACTIVITY

1.2.1: IP addresses, domain names, and URLs.

456048.366176.qx3zqy7

Start

Which IPv4 address(es) are valid?

- (A) ☐ 106.139.254.124
- (B) ☐ 39.34.143.146
- (C) ☐ 363.279.272.30
- (D) ☐ F9.51.09.47

1

2

3

4

Check

Next

View solution ▼ (Instructors only)

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1.3 HTTP

Introduction to HTTP

The **HyperText Transfer Protocol (HTTP)** is a networking protocol that runs over TCP/IP and governs communication between web browsers and web servers. **Transmission Control Protocol/Internet Protocol (TCP/IP)** is a protocol suite that governs how data packets are transferred over the internet from one machine to another. Understanding the details of TCP/IP is not usually required of web developers, but a thorough understanding of HTTP is necessary to create effective web applications.

HTTP versions

HTTP/1.1 is the HTTP standard used for most of the web's lifetime, but many websites are adopting **HTTP/2**, a relatively new HTTP standard that speeds-up the transfer of information between web browsers and web servers. HTTP/2 maintains most of HTTP/1.1's semantics. **HTTP/3**, currently in development, improves the speed of HTTP/2 by using UDP to transport data packets instead of TCP. This material focuses on the basic HTTP workings that all standards share.

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Before HTTP communication begins, the web browser extracts the domain name from the URL being accessed and performs a DNS lookup. The web browser performs a **DNS lookup** by sending the domain name to the local DNS and getting back the IP address of the web server hosting the domain name. Ex: <https://www.w3c.org/> has a domain name of w3c.org, and DNS translates w3c.org to the IP address 193.51.208.66. The web browser uses the IP address to establish a TCP connection with the web server and begins communicating with HTTP.

HTTP functions as a request-response protocol between web browsers and web servers:

- An **HTTP request** is a message sent from the web browser to the web server. Often the request asks the web server to send back a web resource like an HTML file, image, CSS stylesheet, JavaScript file, or video.
- An **HTTP response** is a message sent from the web server back to the web browser in response to an HTTP request. Often the response contains the requested web resource.

**PARTICIPATION
ACTIVITY**

1.3.1: HTTP requests and responses.

**Animation content:**

The web browser with the URL <http://w3c.org/test.html> is displayed. A web server is also displayed. A TCP connection connects the web browser and the web server. The web browser sends the HTTP request asking for test.html to the web server. The web server then sends the HTTP response with the contents `<html>  </html>` back to the web browser. The web browser then sends a HTTP request for field.jpg and the web response then sends back the HTTP response with the contents of the image back to the web browser. Since there are no more requests, the TCP connection is then closed.

Animation captions:

1. The web browser does a DNS lookup of the domain name w3c.org and uses the IP address to create a TCP connection to the web server.
2. The web browser sends an HTTP request asking for the resource test.html.
3. The web server returns the contents of test.html to the browser in an HTTP response.
4. The browser parses the HTML and sends a second HTTP request for field.jpg.
5. The web server returns the image to the browser in an HTTP response. The browser displays the image in the webpage.
6. When no more resources are needed, the browser closes the TCP connection.

**PARTICIPATION
ACTIVITY**

1.3.2: HTTP requests and responses.



- 1) Before a TCP connection is created, the web browser performs a/an _____.
 - ☐ HTTP request
 - ☐ HTTP response
 - ☐ DNS lookup
- 2) How many total HTTP requests does a browser send for a webpage that does not use any other web resources?
 - ☐ 1
 - ☐ 2
 - ☐ 3



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3) How many total HTTP requests does a browser send for a webpage that contains four web resources: an image, a video, a CSS stylesheet, and a JavaScript file?

- ☐ 1
☐ 4
☐ 5

4) If a web browser sends 20 HTTP requests to a web server, how many HTTP responses will the web server likely send to the web browser?

- ☐ 0
☐ 10
☐ 20



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Request and response headers

An HTTP request and an HTTP response are both composed of four parts:

1. Start line - The **start line** specifies the HTTP version being used. A request's start line includes a request type and path; a response's start line includes a status code and phrase.
2. Zero or more header fields - A **header field** is a keyword followed by a colon and a value. Header fields supply additional information about the request or response.
3. Empty line
4. Optional message body - A **message body** contains data being transferred between a web browser and web server. In a request, the message body may be empty or contain submitted form data. In a response, the message body may contain the requested resource.

The figures below show an example HTTP request for a Wikipedia URL and the corresponding HTTP response containing HTML.

Figure 1.3.1: HTTP request with no message body.

Request for URL: https://en.wikipedia.org/wiki/World_Wide_Web

```
GET /wiki/World_Wide_Web HTTP/1.1
Host: en.wikipedia.org
User-Agent: Mozilla/5.0 Chrome/48.0.2564
```



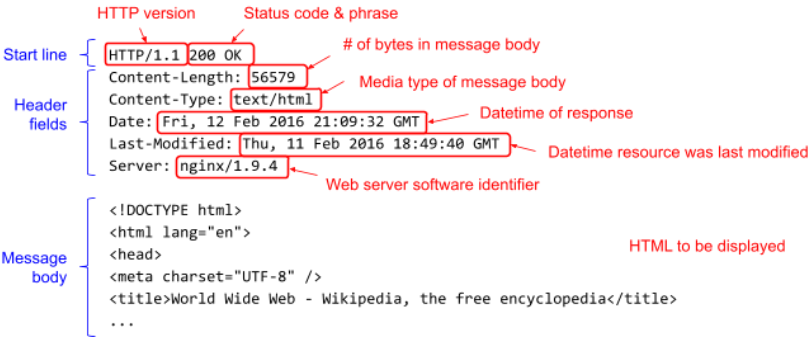
Figure 1.3.2: HTTP response with HTML in message body.

Response for URL: https://en.wikipedia.org/wiki/World_Wide_Web

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```
HTTP/1.1 200 OK
Content-Length: 56579
Content-Type: text/html
Date: Fri, 12 Feb 2016 21:09:32 GMT
Last-Modified: Thu, 11 Feb 2016 18:49:40 GMT
Server: nginx/1.9.4

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<title>World Wide Web - Wikipedia, the free encyclopedia</title>
...
```



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PARTICIPATION
ACTIVITY

1.3.3: HTTP request and response headers.

If unable to drag and drop, refresh the page.

Host

User-Agent

Date

Server

Content-Type

Content-Length

Last-Modified

	Number of bytes in the response's message body.
	Media type of the response's message body.
	Datetime the response was generated by the web server.
	Datetime the requested resource was last modified on the web server.
	Identifies the web server software that generated the response.
	The domain name for the requested path.
	Identifies the browser making the request.

Reset

List of HTTP headers

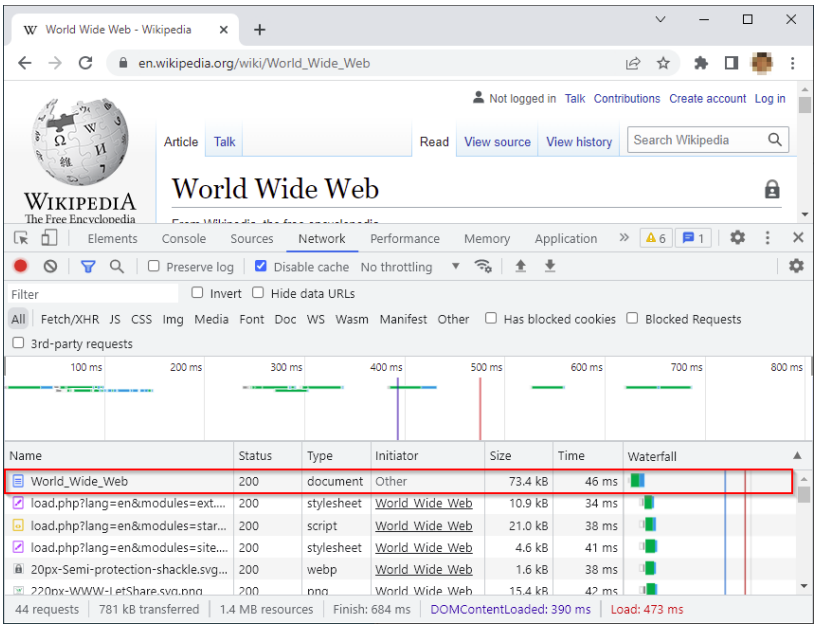
The **Internet Assigned Numbers Authority (IANA)** is a standards organization that manages various internet numbers and symbols, like global IP address allocation, root

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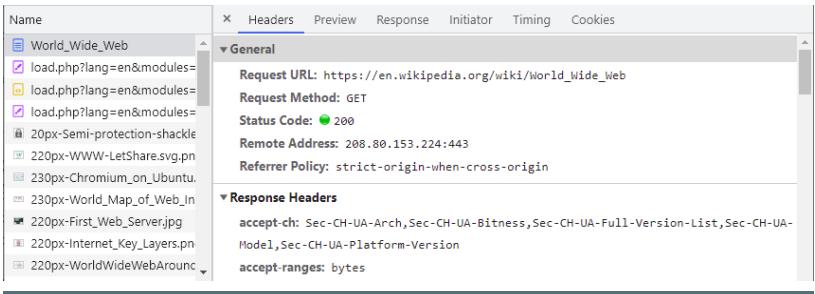
zone management in DNS, and media types. IANA maintains a [list of HTTP headers](#) that are currently active, obsolete, or experimental.

Chrome DevTools for watching HTTP traffic.

All popular web browsers contain built-in developer tools. Developers can access **Chrome's developer tools (DevTools)** by pressing Ctrl+Shift+I (Windows) or Command-Option-I (Mac). In the figure below, the Network tab shows the HTTP network traffic when accessing Wikipedia's article on World Wide Web. The first HTTP request is highlighted. All subsequent requests are for other resources used in the webpage.



Clicking on a specific request shows the request and response details.



Request methods and response status codes

The most common HTTP request-response scenario is when the web browser issues a request with the GET request method, and the web server returns the requested resource with a 200 status code. An HTTP **request method** indicates the desired action to perform on a resource. Other request methods besides GET may also be sent in an HTTP request. Ex: POST is often used when the web browser is sending information from a web form to the web server. The POST, PATCH, PUT, and DELETE request methods are used by web services that allow resources to be created, modified, and deleted on the web server.

Table 1.3.1: Common HTTP request methods.

Request method	Meaning
GET	Request a representation of the specified resource.
HEAD	Request a response identical to GET but without the response body.
POST	Create a new resource with the contents of the message body.
PATCH	Modify an existing resource with the contents of the message body.
PUT	Replace an existing resource with the contents of the message body.
DELETE	Delete an existing resource.

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An HTTP response **status code** is a three digit number that indicates the status of the requested resource. A successfully requested resource results in a 200 status code, and other status codes are returned for various reasons. Ex: A 301 or 302 status code redirects the browser to a different URL. A **browser redirect** is when the web server returns a 301 or 302 status code with a **Location** header indicating the URL the browser should load next.

Table 1.3.2: Common HTTP response status codes.

Status code	Status phrase	Meaning
200	OK	Standard response for a successful request.
301	Moved Permanently	The resource should always be requested at a different URL.
302	Found	The resource should temporarily be requested at a different URL.
304	Not Modified	The resource has not been modified since the last time the resource was requested.
403	Forbidden	The web browser does not have permission to access the resource.
404	Not Found	The resource could not be located.
500	Internal Server Error	Something unexpected happened on the web server.

URL shortening

URL shortening is a technique to create shorter URLs that redirect to longer URLs. Ex: http://en.wikipedia.org/wiki/URL_shortening has a short URL of <http://tinyurl.com/urlwiki>. Short URLs are convenient for sharing on social media, especially on Twitter where the number of characters in a post is limited. Common URL shortening services include bit.ly, rebrandly.com, and tinyurl.com.

When a user types or clicks on a short URL, the URL shortening service responds with a 301 status code and a Location header with the webpage's full URL. The example HTTP request and response below shows <http://tinyurl.com/urlwiki> redirects to http://en.wikipedia.org/wiki/URL_shortening.

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HTTP request	HTTP response
GET /urlwiki HTTP/1.1 Host: tinyurl.com User-Agent: Mozilla/5.0 Chrome/48.0.2564	HTTP/1.1 301 Moved Permanently Date: Tue, 16 Feb 2016 16:38:59 GMT Location: http://en.wikipedia.org/wiki/URL_shortening

**PARTICIPATION
ACTIVITY**

1.3.4: HTTP request methods and response status codes.

- Which request method is used by the browser to get the same response headers that a GET would generate but without the message body?
 - ☐ GET
 - ☐ HEAD
 - ☐ PUT
- Which request method is used by the browser to submit web form data to the web server?
 - ☐ GET
 - ☐ PUT
 - ☐ POST
- Which status code is sent when the web server wants to redirect the browser to a different URL?
 - ☐ 200
 - ☐ 301 or 302
 - ☐ 403
- Which status code is sent when the requested URL does not point to an existing resource on the web server?
 - ☐ 200
 - ☐ 301
 - ☐ 404
- Which status code is sent when the web browser is denied permission to the requested URL?
 - ☐ 301
 - ☐ 304
 - ☐ 403

Browser caching

Most web browsers use a browser cache to store requested content. A **browser cache** is an area on the computer's file system where web content can be stored by the web browser for quick retrieval later. By caching web content, browsers can reduce the amount of network traffic required to display previously visited webpages. Ex: If a webpage is accessed now and again ten minutes from now, the browser can display the cached webpage instead of re-downloading the webpage. If the webpage has changed in the 10 minute span, the browser should download the updated page.

Web browsers often use ETags to aid in caching web resources. An **entity tag (ETag)** is an identifier for a specific version of a web resource. Ex: 34905a3e285dd11. When the resource changes, so should the ETag associated with the resource. When a web browser requests a cached web resource, the browser sends the ETag in the request with an **If-None-Match** header. The web server will reply with a 304 Not Modified response status if the resource has not changed or a 200 OK with the changed resource and a new ETag.

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1.3.5: Requesting cached resources with ETags.

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Animation content:

The web browser with the url `http://w3c.org/test.html` is displayed. A web server is also displayed with the resource `test.html` and an `Etag:123abc`. A get request is sent from the web browser to the web server requesting `test.html`. The web server returns the contents of `test.html` which includes `200 OK, Etag:123abc` back to the browser. These contents are stored inside the browser cache. The web browser sends another request for `test.html` which includes the `Etag:123abc`. A `304 Not Modified` [no message body] is sent back to the web browser. The browser then displays the cached page. The web browser then makes another request for `test.html` with the `Etag 123abc`. The web server compares the ETags and then returns a `200 OK, Etag: 678efg` back to the web browser. The web browser now stores the returned content inside the browser cache and displays it.

Animation captions:

1. The browser requests uncached resource `test.html`. The web server returns the contents of `test.html` back to the browser where `test.html` is cached.
2. The second request for cached `test.html` includes the `Etag 123abc`.
3. The web server compares ETags and notes the file has not changed, so `304` is returned with no HTML. The browser displays the cached content.
4. `test.html` is modified on the web server and assigned a new ETag.
5. The browser sends a third request for `test.html` with `Etag 123abc`.
6. The web server compares ETags and notes the file has changed, so `200` is returned with new HTML and ETag.
7. The browser displays the new content and updates the browser cache.

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**PARTICIPATION
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1.3.6: Requesting cached content.

- 1) Which status code is sent when the web server compares the cached ETag with the server's ETag for a requested web resource, and the two ETags are identical?
 - ☐ 200
 - ☐ 304
 - ☐ 403
- 2) Does a `304` response generally contain a message body?
 - ☐ Yes
 - ☐ No
- 3) Do web servers always generate ETags for all web resources?
 - ☐ Yes
 - ☐ No
- 4) Where can the HTTP header field **If-None-Match** be found?
 - ☐ HTTP request
 - ☐ HTTP response

Other caching mechanisms

HTTP defines additional HTTP headers to aid in caching:

- **If-Modified-Since** is used with the **Last-Modified** date/time to request the web server only send the requested resource if the resource has changed since the specified date/time. Ex: "If-Modified-Since: Wed, 01 Sep 2019 13:24:52 GMT" asks the web server to send the resource if the resource was modified after Sep 1, 2019 at 13:24:52 GMT.

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- **Expires** contains a date/time indicating when the requested resource is considered "stale". Ex: "Expires: Wed, 01 Sep 2019 13:24:52 GMT" tells the web browser to show the cached resource until Sep 1, 2019 at 13:24:52 GMT.
- **Cache-Control** is used to specify a number of caching directives. Ex: "Cache-Control: no-store" tells the web browser to never cache the requested resource, and "Cache-Control: max-age=180" tells the browser to cache the resource for 180 seconds.

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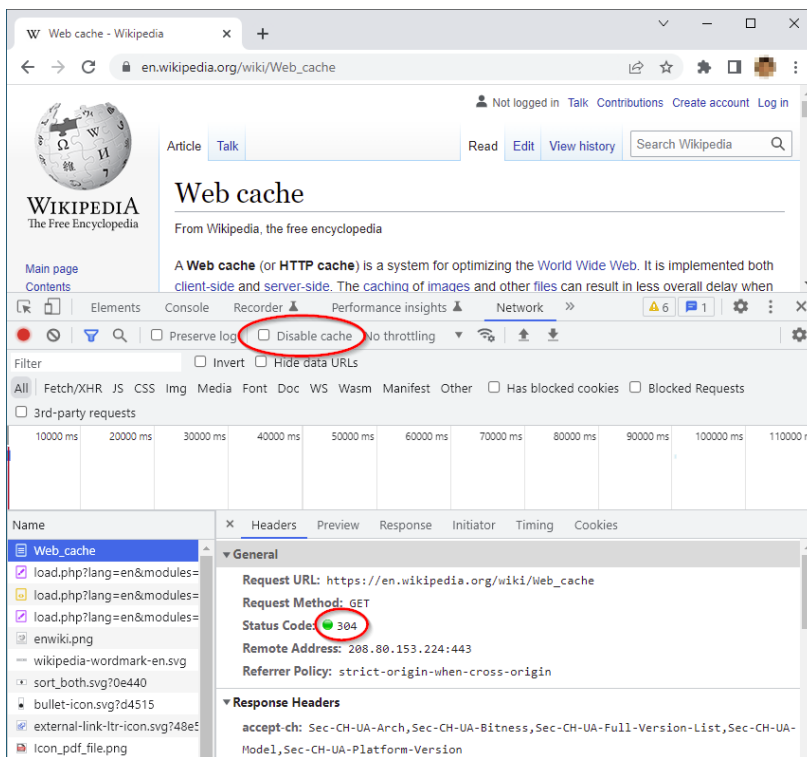
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Viewing 304 responses in Chrome DevTools

Web developers can view 304 Not Modified responses in Chrome's DevTools by opening the Network tab and ensuring the checkbox labeled "Disable cache" is not checked. If "Disable cache" is checked then nothing will be cached, and **If-None-Match** and **If-Modified-Since** headers will not be sent.

The Wikipedia page about [web cache](#) is a good page to test in DevTools to see 304 responses. When the developer loads the page once, the web server returns a 200 status code for the Web_cache request. When the developer presses the browser's Reload button, the server returns a 304 status code.

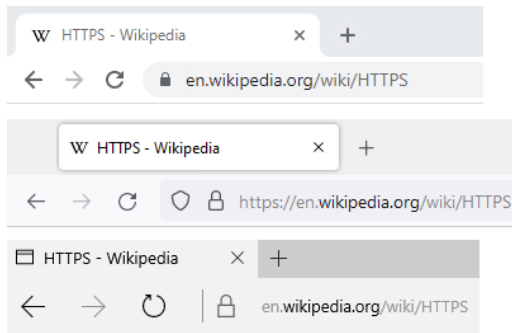


HTTPS

All HTTP traffic can be viewed by third parties using a network sniffer. A **network sniffer** is software that monitors network traffic and allows users to inspect HTTP requests and responses. **HTTPS** encrypts HTTP traffic between a browser and web server so a network sniffer cannot intercept sensitive information in the HTTP traffic like passwords, credit card numbers, financial transactions, etc.

HTTPS uses a protocol called **Transport Layer Security (TLS)**, which uses asymmetric public keys to encrypt data between the browser and web server. A website wanting to use HTTPS must acquire a **digital certificate**, issued by a trusted **certificate authority**, that contains a public key used by TLS to encrypt data.

Figure 1.3.3: Chrome, Firefox, and Edge browsers showing a padlock symbol when HTTPS is used.



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1.3.7: Steps in an HTTPS transaction.

Put the steps used in an HTTPS transaction in order.

If unable to drag and drop, refresh the page.

The browser and web server transmit encrypted information.

Web server sends digital certificate to the browser.

The browser and web server initiate "TLS handshake".

Browser requests an HTTPS connection to a webpage.

	Step 1
	Step 2
	Step 3
	Step 4

Reset

Exploring further:

- [Anatomy of an HTTP Transaction](#)
- [HTTP response status codes](#) (MDN)
- [ETag - HTTP](#) (MDN)
- [Moving to HTTPS Guide](#)
- [Inspect Network Activity with Chrome DevTools](#)
- [HTTP/2](#)
- [HTTP/3](#)

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**CHALLENGE
ACTIVITY**

1.3.1: HTTP.

456048.366176.qx3zqy7

Start

For each webpage described below, indicate the total HTTP requests the web browser

sends and the total HTTP responses the web server sends, assuming that no requests are ignored.

	Webpage contents	HTTP requests	HTTP responses
1	2 JavaScript files, 2 videos, and an image	Ex: 9	
2	3 JavaScript files and an image		
3	2 videos		

1

2

3

4

Check

Next

View solution (Instructors only)

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1.4 Web trends

Web technology is changing so fast, that predicting how web tools and behaviors will have changed, even a few years from now, is very challenging. However, some significant trends exist and are important for web developers to consider.

Making predictions

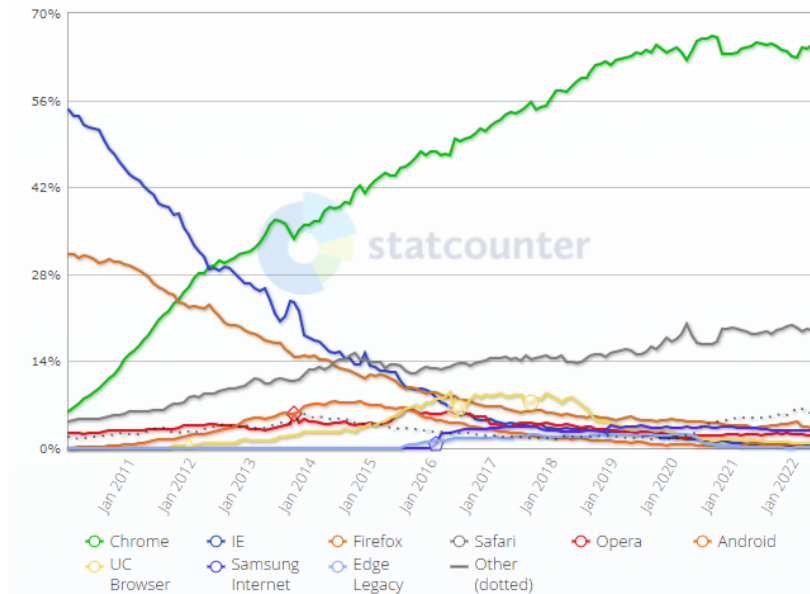
Yogi Berra, the famous baseball player and manager, joked "It's tough to make predictions, especially about the future."

Web browser market share continues to change

The relative usage of various browsers changes continually. Web applications that depend on a particular browser to function correctly can be frustrating to users. *Good practice is to ensure that webpages adhere closely to current technology standards, and that webpages work acceptably on all established browsers.*

Figure 1.4.1: Global browser usage trends.

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PARTICIPATION ACTIVITY

1.4.1: Implications of the market share browsers.

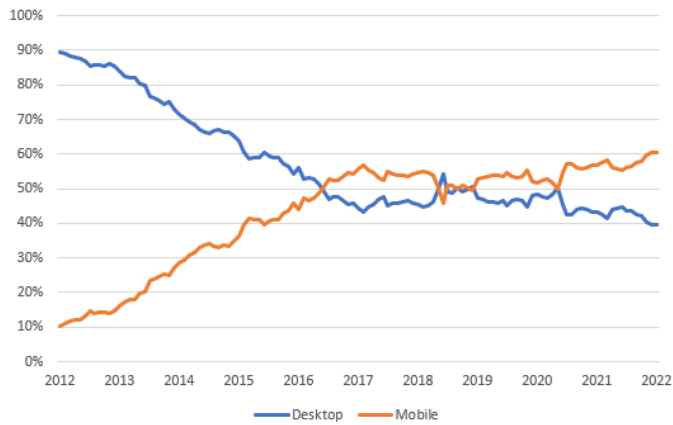
- 1) Because Chrome is by far the leading browser worldwide, testing a webpage only really needs to be done using Chrome.
 - ☐ True
 - ☐ False
- 2) Web developers typically create webpages using new features available exclusively on a particular browser.
 - ☐ True
 - ☐ False
- 3) Web developers can sometimes ignore really old web browsers.
 - ☐ True
 - ☐ False

Mobile devices are replacing desktop computers on the web

A **mobile device** is a handheld computer, like a smartphone or tablet. Mobile devices are used to access the web today more frequently than desktop computers. Web developers need to design applications that work equally well on desktop and mobile devices.

Figure 1.4.2: Global browser usage trends: Desktop vs. mobile.

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Data source: [StatCounter GlobalStats](#) (June 2012 to May 2022)

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Developers of mobile-friendly webpages need to consider the following:

- Screen size: Screen size is much smaller than desktops.
- Load speed: Mobile devices may have limited or slower internet connectivity.
- Device speed: Limited memory and CPU speed of mobile devices means mobile browsers are not as powerful as desktop browsers.
- Data cost: Many users have data plans that limit how much content can be downloaded. Large webpages and webpages that keep requesting more data reach a mobile data plan's limit faster.
- Battery life: A webpage that constantly runs JavaScript and frequently loads data will drain the battery faster.
- Interface: User interaction is with touch, not a mouse.

PARTICIPATION ACTIVITY

1.4.2: Important mobile development topics.

If unable to drag and drop, refresh the page.

Mobile first

Affordances

Responsive web design

Screen size

A web development approach that advocates first creating a reduced-feature version of a website for mobile users. Then, the developer creates a full-featured website for visitors using desktop computers.

Physical dimension in pixels of a device's screen.

Visual clues that guide the user in figuring out how to use an app.

A web design approach that creates webpages that automatically move and resize parts depending on the display size and orientation.

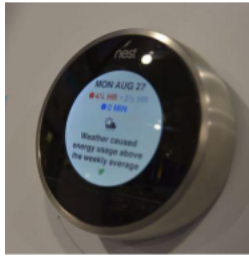
Reset

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IoT: The Internet of Things

The **Internet of Things** (abbreviated as IoT) is the global collection of communicating devices that sense and control technology on behalf of humans. IoT devices range from a simple temperature sensor to a satellite-based laser scanner used to discover archaeological sites hidden by vegetation.

Example 1.4.1: Example Internet of Things devices.



Nest learning thermostat



LG Internet refrigerator

Source: Nest learning thermostat ([Raysonho @ Open Grid Scheduler / Grid Engine](#) / [Public domain via Wikimedia Commons](#)),

Internet refrigerator ([LG전자](#) / [CC-BY-2.0](#) via Wikimedia Commons)

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IoT devices display one or more of the following characteristics:

- Gather information about the physical world using sensors. Ex: temperature, voltage, images.
- Share the sensor data with control systems. Ex: A sensor may report a room's temperature to building control system, or air ozone and water pollution sensors reporting air quality data to urban environment monitoring systems.
- Interact with hardware to execute commands sent by control systems. Ex: Speed up fan moving hot air to part of a building where the sensed temperature is below the target temperature.

People do not typically interact directly with IoT devices. Instead, people access information and control devices using custom apps or webpages. Web developers typically build the webpages that present a system overview and allow users to select and display specific information from the aggregated sensor data. Webpages for IoT may allow a user to change system parameters or activate/deactivate parts of the system.

PARTICIPATION ACTIVITY

1.4.3: Internet of Things.

Which statements are likely true about an IoT-based system used to monitor a swimming pool for a pool service company?

1) A technician installs devices that measure and report water chemical concentrations, water level, pump status, and water flow through the filter system.

- ☐ True
☐ False

2) The pool control system periodically transmits summary data over the internet to a pool service company's web server.

- ☐ True
☐ False

3) The pool company would have to send a technician to the pool to change the control system settings.

- ☐ True
☐ False

4) The pool company will not know that the water level is too low unless the customer calls or someone happens to check the system status webpage.

- ☐ True
☐ False

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5) If the customer notices the low water level and calls, the company receptionist's browser can have the customer's pool status webpage loaded by the time the receptionist answers the phone.

- ☐ True
☐ False

6) After the customer reports the water level, the receptionist could inform the caller that a service technician was already dispatched and was six minutes from arriving at the house.

- ☐ True
☐ False

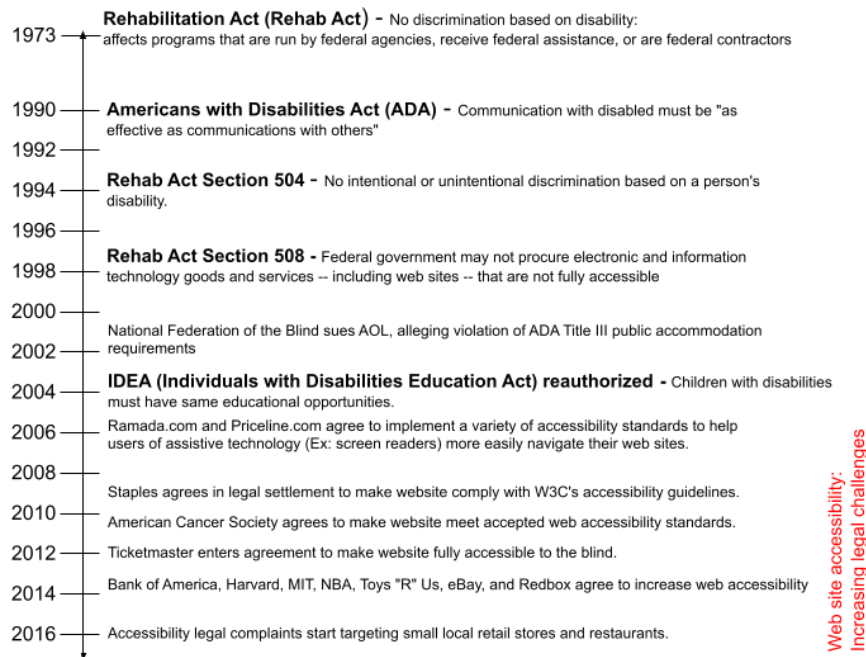
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Web accessibility

Web accessibility is the ability of users with disabilities to access and use a webpage with reasonable effort. Designing accessible webpages ensures equal access and opportunity for everyone. Developing accessible webpages requires knowledge of disabilities, assistive technologies and software used by users with disabilities, and following design practices to ensure content is compatible with those assistive tools. Some conditions affecting web accessibility include:

- Visual problems like blindness, low vision, or color blindness
- Hand control issues ranging from tremors to total inability to use hands
- Seizures caused by flashing on the screen
- Cognitive challenges like dyslexia and other difficulties in processing webpages

Figure 1.4.3: Web accessibility timeline.



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PARTICIPATION ACTIVITY

1.4.4: Accessibility fact check.

1) Really small businesses are exempt from web accessibility requirements.

- ☐ True
☐ False

2) Color blindness is so rare that web developers do not have to consider the issue when selecting colors for a webpage.

- ☐ True
☐ False

3) A random website is more likely to have a visitor with low vision than a visitor who is totally blind.

- ☐ True
☐ False

4) Unlike the blind, people who are deaf do not need special accommodation for websites.

- ☐ True
☐ False



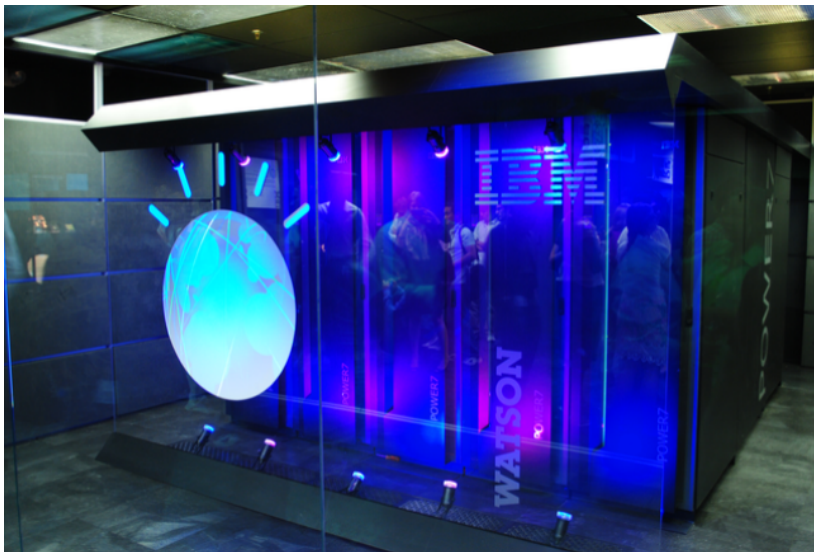
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Cognitive computing and big data intelligence

Cognitive computing is the use of artificial intelligence techniques and access to vast amounts of data to simulate human problem solving in complex situations with ambiguity, changing data, and even conflicting information. IBM Watson is the symbol of this trend to create intelligent software systems that process massive numbers of webpages in order to extract information and address challenging problems in areas such as medical treatment, increasing retail sales, and improving the quality of call center help.

When processing webpages, intelligent systems like Watson are aided by effective use of webpage HTML metadata and structure markup. Ex: Text containing digits is more useful when the digits are placed in an HTML table with a title like "Pacemaker failure rates".

Figure 1.4.4: IBM Watson cognitive computing system.



Source: IBM Watson([Clockready](#) / [CC-BY-SA-3.0](#) via Wikimedia Commons)

Separation of concerns

Separation of concerns is the design principle of breaking up web content using distinct languages and documents that overlap as little as possible. In modern web design, document structure and text, visual layout, and page interaction are separately specified using three key languages.

- HTML is the language that describes the page content.
- CSS is the language that describes page layout.
- JavaScript is the language that provides interactive functionality.

When webpages are built so as to cleanly separate document structure, visual layout, and interaction, the following results are obtained:

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- Webpages are more likely to work as intended across a wide range of browsers.
- Webpages work better on smartphones and other mobile devices.
- Internet of Things systems are easier to build because relevant data is easier to identify.
- Web accessibility is improved.
- Intelligent systems can extract more meaning from the content of webpages.

Web developers are expected to deliver content that will work well with all devices and browsers. Ideally, a web developer builds websites that will work on any browser, but can take advantage of recently added features available in newer browsers. Ex: A website that adds additional functionality for touch interfaces on mobile devices, but remains functional on desktops without a touch interface. As the variety of devices continues to proliferate, developing websites that comply to web standards ensures the website will operate correctly on as many devices as possible.

PARTICIPATION ACTIVITY

1.4.5: Review trends in web technology.

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1) Which browser lost the most market share between 2010 and 2019?

- ☐ Safari
- ☐ Chrome
- ☐ IE (Internet Explorer)

2) In what year did web browsing first become more likely on mobile rather than desktop?

- ☐ 2010
- ☐ 2016
- ☐ 2018

3) Are human doctors or IBM's Watson more likely to keep up with new medical information in the future?

- ☐ Human doctors
- ☐ IBM's Watson

Exploring further:

- [Internet of Things article in McKinsey Quarterly.](#)
- [W3C Web Accessibility initiative document](#)
- [Overview of United States laws relevant to web accessibility.](#)

1.5 Introduction to HTML

HTML and tags

HTML, CSS, and JavaScript are used together to create a webpage.

- HTML defines the structure and content of a webpage.
- CSS specifies the layout and visible appearance.
- JavaScript describes the dynamic behaviors and actions of a webpage.

As an analogy, humans have similar components: structure (bones, organs, central nervous system), identifying attributes (eye color, hair style, height), and behaviors (brushing teeth, slam dunking a basketball).

HTML (HyperText Markup Language) is a textual language for creating webpages. HTML files are usually saved with a .html or .htm file extension. Ex: index.html. An HTML file starts with an indication of the document type, then a head part with the page title and other page information, and finally a body part with the actual page content. The HTML file may also contain CSS and JavaScript code.

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Figure 1.5.1: An HTML file and the rendered webpage.


```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>For sale: 2012 Ducati Streetfighter</title>
  </head>
  <body>
    
    <h1>2012 Ducati Streetfighter 848 - $9000</h1>
    <p>2012 Ducati Streetfighter 848, Low Miles, Lots of Upgrades. Full service history. Call or text 555-4400.</p>
    <p>year: <strong>2012</strong></p>
    <p>make and model: <strong>Ducati Streetfighter 848</strong></p>
    <p>condition: <strong>excellent</strong></p>
    <p>engine: <strong>848</strong></p>
    <p>odometer: <strong>9500</strong></p>
    <p>paint color: <strong>yellow</strong></p>
  </body>
</html>
```

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2012 Ducati Streetfighter 848 - \$9000

2012 Ducati Streetfighter 848, Low Miles, Lots of Upgrades. Full service history. Call or text 555-4400.

year: **2012**

make and model: **Ducati Streetfighter 848**

condition: **excellent**

engine: **848**

odometer: **9500**

paint color: **yellow**

Source: zyBooks

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An HTML document is constructed with elements. An **element** is a single HTML structure that is represented with HTML tags.

A **tag** has a descriptive name surrounded by < and > characters that the web browser uses to display content. Ex: <p> specifies a paragraph. Most HTML elements have an opening and closing tag. Ex: always has an opening tag and closing tag . Some tags, like , do not require a closing tag.

Table 1.5.1: Common HTML elements.

Elements	Purpose	Example HTML
h1, h2, h3	Headers. h1 is largest.	Puppies are cute

p	Paragraph	<code><p>Humans seem designed to see puppies as cute.</p></code>
em, strong	Emphasis, strong emphasis	<code><p>"When in doubt, tell the truth" - Mark Twain.</p></code>
img	Image	<code></code> Note: <code>src</code> is the image's URL source, and <code>alt</code> describes the image.
a	Link	<code>Click Here</code>

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PARTICIPATION
ACTIVITY

1.5.1: Practice with HTML tags.

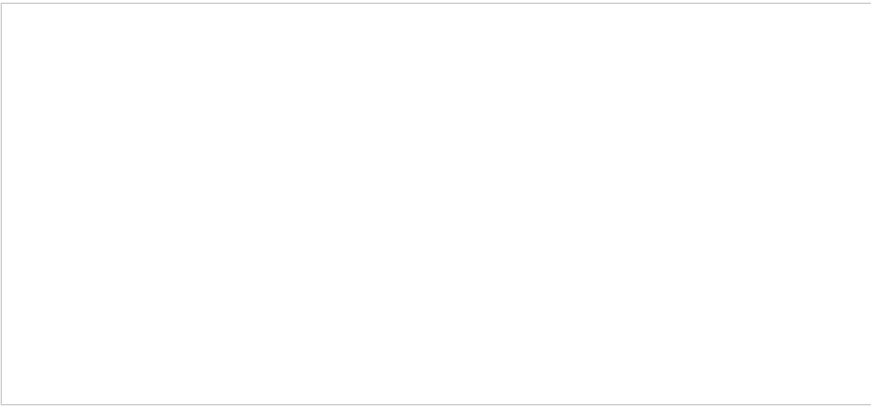
Modify the HTML to include h2 and h3 headers, bold text, another paragraph, etc.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <title>What I want to know...</title>
6   </head>
7   <body>
8     
9     <h1>Is Pluto a Planet?</h1>
10
11     <p>When I was young, <em>Pluto</em> was a planet. Then later, Pluto was
12     Now, Pluto may be a planet again. Well, <strong><em>is it or isn't it??</em></strong>
13
14   </body>
15 </html>
16
```

Render webpage

Reset code

Your webpage



View solution  (Instructors only)

PARTICIPATION
ACTIVITY

1.5.2: Match HTML basic formatting.

If unable to drag and drop, refresh the page.

`ABC``<h3>ABC</h3>``<p>ABC</p>``ABC`

`<strong>ABC</strong>`

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	ABC
	ABC
	ABC
	ABC
	ABC

[Reset](#)

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Links

A **link** on a webpage is a clickable item that usually causes the web browser to open another webpage when clicked. Ex: [Dear Photograph](#) is a link to an interesting website. A button or image can also be a link.

A link is created with an anchor element. The opening `<a>` tag uses the `href` attribute to specify the link's URL target. The link text comes next and is followed by the closing `</a>` tag.

PARTICIPATION ACTIVITY

1.5.3: Creating a link in HTML.

Change the link's URL to your favorite webpage. Then add a second link to the words "second favorite webpage". After clicking a link in the rendered webpage, re-render the page to see the links again.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>My Favorite Webpages</title>
5   </head>
6   <body>
7     <p>Visit my <a href="https://www.wikipedia.org/">favorite webpage</a></p>
8     <p>Visit my second favorite webpage.</p>
9   </body>
10 </html>
11
```

[Render webpage](#)[Reset code](#)

Your webpage

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PARTICIPATION ACTIVITY

1.5.4: HTML links.

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Are the following valid HTML links?

- 1) `<a "https://cnn.com">CNN</a>`

☐ Valid

☐ Not valid
- 2) `<a href="https://cnn.com">News</a>`

☐ Valid

☐ Not valid
- 3) `<a href="https://www.youtube.com/watch?v=uu7XCeMdSHg">News</a>`

☐ Valid

☐ Not valid

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CHALLENGE ACTIVITY

1.5.1: Introduction to HTML.

456049.366176.qx3zqy7

Start

Before the paragraph, add an h2 header that reads 'Introduction' and then an h3 header that reads 'References'. [SHOW EXPECTED](#)

```

1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <title>My first html program with headers</title>
6   </head>
7   <body>
8
9     <!-- Your solution goes here -->
10
11     <p>Here are some references</p>
12   </body>
13 </html>

```

1

2

3

4

5

Check

Next

View solution ▼ (Instructors only)

1.6 Introduction to CSS

CSS and rules

A webpage without any styling will use the browser's default styling with white background and black, standard-sized text. **Cascading Style Sheets (CSS)** is a textual language for describing how a webpage is styled for visual presentation. CSS controls the look and layout of webpage content.

A **CSS rule** specifies styling properties for specific HTML elements. CSS rules may be placed within `<style>` tags in the HTML file's head part. Each rule indicates the element to be styled like h1 (header1) or p (paragraph), followed by a list in braces { } of property:value items like `color:blue`.

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Figure 1.6.1: HTML code without and with CSS rules.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>For sale: 2012 Ducati
Streetfighter</title>
  </head>
  <body>
    <h1>Ducati Streetfighter -
$9000</h1>
    <p>year: <strong>2012</strong></p>
    <p>make and model:
<strong>Ducati Streetfighter
848</strong></p>
    <p>condition:
<strong>excellent</strong></p>
    <p>odometer: <strong>9500</strong>
</p>
  </body>
</html>
```

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>For sale: 2012 Ducati
Streetfighter</title>
    <style>
      h1 {
        color: green;
        background-color: lightgray;
      }
      p {
        font-family: arial;
        margin-left: 10px;
      }
      strong {
        background-color: lightgreen;
        padding: 5px;
      }
    </style>
  </head>
  <body>
    <h1>Ducati Streetfighter -
$9000</h1>
    <p>year: <strong>2012</strong></p>
    <p>make and model:
<strong>Ducati Streetfighter
848</strong></p>
    <p>condition:
<strong>excellent</strong></p>
    <p>odometer: <strong>9500</strong>
</p>
  </body>
</html>
```

Ducati Streetfighter - \$9000

year: 2012
make and model: **Ducati Streetfighter 848**
condition: **excellent**
odometer: **9500**

Ducati Streetfighter - \$9000

year: **2012**
make and model: **Ducati Streetfighter 848**
condition: **excellent**
odometer: **9500**

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Table 1.6.1: Common CSS properties.

Properties	Purpose	Example CSS
background-color	Element's background color	<pre>h1 { background-color: lightgray; }</pre>
color	Element's font color	<pre>h1 { color: darkblue; }</pre>
font-family	Font used for the element's text.	<pre>p { font-family: arial; }</pre>
font-size	Font size used for the element's text.	<pre>p { font-size: 9pt; }</pre>
padding	Spacing between element's content and border.	<pre>strong { padding: 5px; }</pre>

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margin	Spacing around element (outside element's border).	<pre>p { margin-left: 10px; }</pre> Note: margin-top, margin-bottom, margin-left, and margin-right specify margin for individual sides.
--------	--	--

PARTICIPATION
ACTIVITY

1.6.1: Styling a webpage with CSS.

Try changing the h1's font size to 16pt, the p's left margin to 30px, and the p's font family to Helvetica.

```
1 <!DOCTYPE html>  
2 <html lang="en">  
3   <head>  
4     <meta charset="UTF-8">  
5     <title>For sale: 2012 Ducati Streetfighter</title>  
6   <style>  
7     h1 {  
8       color: green;  
9       background-color: lightgray;  
10    }  
11    p {  
12      font-family: georgia;  
13      margin-left: 10px;  
14    }  
15    strong {  
16      background-color: lightgreen;  
17      padding: 5px;  
18    }
```

Render webpage

Reset code

Your webpage

Expected webpage

--	--

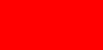
View solution  (Instructors only)

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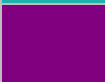
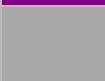
Colors

A **CSS color** can be a pre-defined name like blue, or an rgb value like rgb(50, 100, 255). **rgb** is short for red, green, blue; all colors can be formed by combining those three colors. Values range from 0 (none) to 255 (bright).

Table 1.6.2: Some CSS colors.

Name	RGB	Color	Description
Red	rgb(255, 0, 0)		Bright red
Green	rgb(0, 255, 0)		Bright green
Blue	rgb(0, 0, 255)		Bright blue

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LightSeaGreen	rgb(32, 178, 170)		A particular mix of red, green, and blue, yielding sea green
Purple	rgb(128, 0, 128)		A particular mix of red and blue, yielding purple
DarkGray	rgb(169, 169, 169)		Equal red, green, blue yields gray
White	rgb(255, 255, 255)		Full red, green, blue yields white
Black	rgb(0, 0, 0)		No red, green, or blue yields black

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Color picker.

rgb(0 , 0 , 0)
Each value ranges 0 to 255

#000000
Value ranges 000000 to FFFFFF

Red

Green

Blue



PARTICIPATION
ACTIVITY

1.6.2: CSS colors.

Use the color picker above to determine the color of each rgb value. The hexadecimal value in the upper-right corner of the color picker is equivalent to the rgb value.

If unable to drag and drop, refresh the page.

White

Fuchsia

Orange

Dark red

	rgb(200, 0, 0)
	rgb(255, 0, 255)
	rgb(250, 150, 0)
	rgb(255, 255, 255)

Reset

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PARTICIPATION
ACTIVITY

1.6.3: CSS colors.

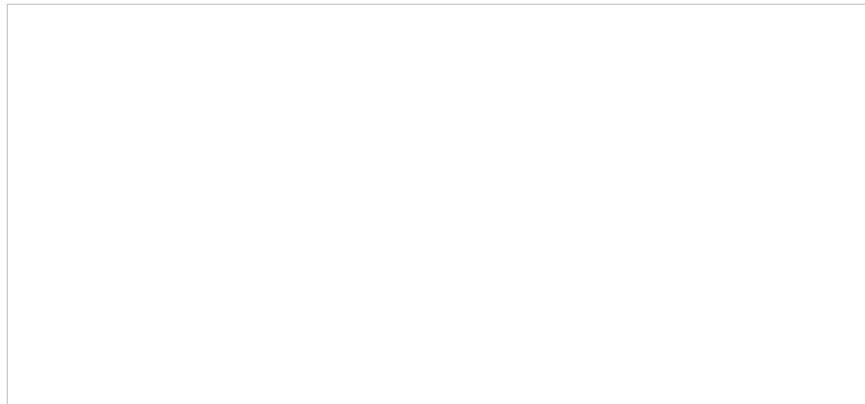
Try changing h1's color to rgb(40,40,40), and changing strong's color to rgb(100,100,100).

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Kyoto Kaiseki Restaurant Review</title>
5     <meta charset="UTF-8">
6     <style>
7       h1 {
8         color: black;
9         background-color: lightgreen;
10      }
11      p {
12        font-size: 12pt;
13        margin-left: 10px;
14      }
15      img {
16        margin-left: 10px;
17      }
18    </style>
19  </head>
```

Render webpage

[Reset code](#)

Your webpage

[View solution](#) (Instructors only)CHALLENGE
ACTIVITY

1.6.1: Introduction to CSS.



456048.366176.qx3zqy7

Start

Change em's font-size to 18pt. [SHOW EXPECTED](#)

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <title>Kyoto Kaiseki Restaurant Review</title>
6     <style>
7       /* Your solution goes here */
8
9
10    </style>
11  </head>
12  <body>
13    <h1>Kyoto Kaiseki Restaurant Review</h1>
14    <h2>Kyoto, Japan</h2>
15    <p><strong>Rating (1-5):</strong> 5</p>
16    <p><strong>Favorite dish:</strong> Mixed sashimi</p>
17
18    
```

1

2

3

Check

Next

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1.7 Introduction to JavaScript

Button click example

JavaScript is a programming language that runs in a browser, enabling webpages supporting actions like responding to a button click. JavaScript can be included in the HTML file's head or body parts.

PARTICIPATION
ACTIVITY

1.7.1: JavaScript to change colors.



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Click the buttons. Try adding a third button for "blue".

```

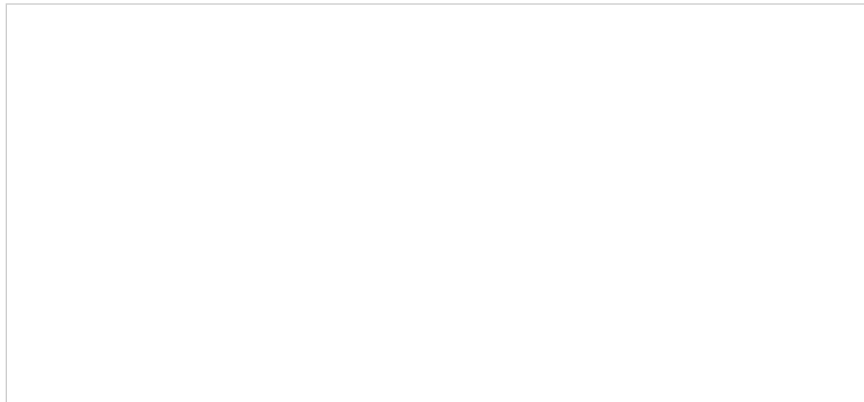
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <title>For sale: 2012 Ducati Streetfighter</title>
6     <style>
7       h1 {
8         color: green;
9         background-color: lightgray;
10      }
11      p {
12        font-family: arial;
13        margin-left: 10px;
14      }
15      strong {
16        background-color: lightgreen;
17        padding: 5px;
18    }

```

Render webpage

Reset code

Your webpage



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In the HTML above:

- The h1 heading has an id of bike-heading. An **id** attribute identifies an HTML element so JavaScript can manipulate the element. Two button elements are also created, each with a different id.
- The **<script>** tags, located near the end of the HTML, surround the JavaScript code. The code `document.getElementById("idName")` searches the HTML document for and returns an element with **id="idName"**. `whiteBtn`, `greenBtn`, and heading are examples of variables. A **variable** stores a value or a link to an element of a webpage. Ex: `whiteBtn` stores the element with `id="white-btn"`.
- The code `buttonVariable.addEventListener("click", function)` specifies code that executes when the button is clicked. Each button calls the function `changeTextColor()` using the value "white" or "green". A JavaScript **function** is a named group of statements that can be executed by referring to that name.
- The `changeTextColor()` function's statements change the h1's color. Assigning the `heading.style.color` with a color like "white" changes the element's CSS, which causes the element's font color to change.

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PARTICIPATION
ACTIVITY

1.7.2: JavaScript for the button click example.



If unable to drag and drop, refresh the page.

<h1 id="bike-heading">

heading = document.getElementById("bike-heading")

<button id="blue-btn">Blue</button>

changeTextColor("white")

Gives the h1 heading a label, so that a function can find the heading to change the heading's color.

Indicates that the JavaScript function changeTextColor() should be executed, with the value "white".

Finds the element with id bike-heading, and sets heading to that element. heading can then be used to change the element's attributes, like color.

The HTML that should be added to create a third button.

Reset

If-else statement example

The JavaScript example below shows a function with an "if-else" statement for setting the color of the rating stars based on the value passed to the updateRating() function. The HTML below defines five span elements, which are inline containers used to manage HTML content. Each span element has a unique id and contains a single * for the rating star. The JavaScript code can change each rating star's color by changing the span's color.

PARTICIPATION ACTIVITY

1.7.3: Updating user ratings.

Click the buttons. Try adding two more buttons for ratings 2 and 1. Try replacing the * with a star (★). To specify a star, use ★, which is the HTML entity for displaying a star.

1<!DOCTYPE html>

2<html lang="en">

3<head>

4<meta charset="UTF-8">

5<title>Kyoto Kaiseki Restaurant Review</title>

6<style>

7h1 {

8background-color: lightgreen;

9color: rgb(40,40,40);

10}

11p {

12font-family: arial;

13margin-left: 10px;

14}

15img {

16margin-left: 10px;

17}

18strong {

Render webpage

Reset code

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PARTICIPATION
ACTIVITY

1.7.4: JavaScript for updating user ratings.

Refer to the JavaScript example above

1) What is the id of the span containing the third rating star?

Check

Show answer

2) If the user clicks the "Rate 4" button, to what color is the fourth rating star set?

Check

Show answer

Drawing graphics example

More advanced interactive webpages, such as a user-entry form or a browser-based video game, may involve hundreds or thousands of JavaScript statements. JavaScript programs are thus commonly placed in a separate file, typically ending in .js, and linked to in an HTML file's head part.

Here is the popular game Tetris, written in JavaScript: [Tetris](#).

JavaScript example: Analog clock.

JavaScript can also be used to draw graphics. Play around with the clock below by changing time. Note: No changes are needed.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <title>Analog Clock</title>
6     <style>
7       .hour-input {
8         color: blue;
9         margin-right: 5px;
10        width: 30px;
11      }
12
13      .minute-input {
14        color: red;
15        margin-left: 5px;
16        width: 30px;
17      }
18    </style>
```

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**PARTICIPATION
ACTIVITY**

1.7.5: JavaScript clock example.

Refer to the clock example above.

- 1) The clock does not change unless a button is clicked.

☐ True

☐ False
- 2) In what tag does the JavaScript code appear in the HTML?

☐ <style>

☐ <script>
- 3) The JavaScript draws the clock inside what tag?

☐ <canvas>

☐ <input>



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